Week 5 Assignment

Replication, randomization, and blocking

Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas before Week 6's lecture (Monday). See the syllabus for the late policy.

Part 1: Design critique

For each scenario, name (a) the likely experimental unit, and (b) one design flaw:

1. A researcher wants to test whether a new diet increases growth rate in fish. They put 30 fish in one large tank with the new diet, and 30 fish in another tank with the standard diet.
2. A field ecologist wants to know if a pesticide affects pollinator visits. They spray one field and leave a neighboring field unsprayed, then count pollinator visits at 20 flowers in each field.
3. A greenhouse study assigns fertilizer treatments to pots by simply giving the new fertilizer to whichever pots were closest to the watering can.

Part 2: Simulation practice

Using this week's tutorial code as a starting point:

1. Change `n_blocks` to 3 (instead of 6) and re-run the simulation and both models.

Does blocking still reduce the residual variance as much as it did with 6 blocks?

2. In 2–3 sentences: what does this suggest about the trade-off between number of blocks and the benefit of blocking?